

CHOPOK, Slovakia

WMO: 119160

Lat: 48.9433N

Lon: 19.5892E

Elev: 6585

StdP: 11.52

Time Zone: 1.00 (EUC)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-1.5	1.6	-24.9	1.8	16.3	-17.6	2.7	19.1	62.4	15.4	57.0	17.3	22.9	330	1.060

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.9	61.0	53.5	58.4	52.4	56.2	51.3	55.6	59.0	53.9	56.9	52.3	55.2	12.4	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
54.3	80.4	56.7	52.6	75.3	55.2	50.8	70.5	53.7	26.5	59.2	25.2	56.8	24.1	55.5	60.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 52.6	46.8	41.8	-7.1	65.1	4.7	2.7	-10.5	67.0	-13.2	68.5	-15.9	70.0	-19.3	71.9		
(5)			-8.0	57.9	4.6	1.5	-11.3	59.0	-14.0	59.8	-16.6	60.6	-20.0	61.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	31.8	17.5	17.3	20.7	29.3	37.8	44.0	47.2	47.5	40.0	33.4	26.4	19.9
	DBStd	13.39	8.21	8.05	8.04	7.96	7.12	7.09	6.31	6.10	6.67	8.57	8.48	8.48
	HDD50	6745	1008	915	908	621	383	201	130	119	304	516	708	932
	HDD65	12102	1473	1335	1373	1071	844	629	551	541	750	980	1158	1397
	CDD50	117	0	0	0	0	3	22	44	43	4	1	0	0
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	18.7	21.5	20.8	21.1	19.1	16.6	15.3	15.9	14.0	16.5	18.7	22.4	22.8
Precipitation	PrecAvg	37.8	2.2	2.3	2.4	2.6	3.9	4.2	5.0	3.5	3.2	3.1	3.0	2.3
	PrecMax	52.3	6.8	11.1	6.3	6.5	9.1	6.9	9.1	9.1	6.5	5.9	5.8	5.9
	PrecMin	24.6	0.2	0.3	0.4	0.4	1.5	1.3	1.3	0.9	0.7	0.2	0.4	0.6
	PrecStd	6.3	1.5	2.0	1.5	1.4	1.8	1.5	2.2	1.9	1.7	1.6	1.5	1.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.0	37.0	40.4	47.9	56.9	62.5	64.2	65.0	56.9	54.2	50.3	41.8
		MCWB	28.8	28.8	32.1	40.2	48.4	54.1	55.6	55.4	49.8	40.8	39.1	30.2
	2%	DB	34.8	32.9	35.8	44.3	53.0	59.1	60.9	61.0	53.6	49.9	42.3	36.5
		MCWB	28.0	27.8	30.9	38.5	46.4	52.4	53.7	53.6	47.7	40.4	35.5	28.1
	5%	DB	31.4	30.5	33.3	41.4	50.0	56.3	58.1	58.0	51.1	46.7	39.1	33.3
		MCWB	27.9	27.8	30.3	36.9	45.2	51.3	52.8	52.5	46.4	40.6	35.5	28.4
	10%	DB	28.7	28.2	31.3	38.8	47.3	53.7	55.7	55.8	48.7	44.3	36.8	31.1
		MCWB	26.5	26.7	29.4	35.3	43.6	49.5	51.7	51.2	45.2	40.5	34.1	28.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	33.1	32.3	35.2	42.2	50.4	57.2	57.3	57.9	51.6	46.6	42.2	34.2
		MCDB	34.6	33.6	36.5	45.0	54.5	59.6	60.8	60.7	54.6	47.7	45.9	35.8
	2%	WB	30.9	30.3	32.8	39.6	48.0	54.3	55.3	55.7	49.5	44.7	38.7	31.8
		MCDB	32.2	31.7	34.1	42.8	51.0	57.4	58.8	59.2	51.8	46.0	39.6	32.8
	5%	WB	28.7	28.5	31.2	37.8	46.1	52.0	53.8	53.7	47.8	43.0	36.6	30.4
		MCDB	30.2	29.4	32.4	40.2	48.8	54.9	56.8	56.6	49.8	44.8	37.7	31.9
	10%	WB	26.6	26.5	29.5	36.1	44.1	50.2	52.2	52.0	46.0	41.3	34.7	28.7
		MCDB	28.3	27.7	31.0	38.3	46.8	52.9	55.2	55.0	47.9	43.1	36.2	30.2
Mean Daily Temperature Range	5% DB	MDBR	8.0	8.1	8.2	7.7	7.6	8.0	8.0	7.9	7.2	7.3	7.1	7.7
		MCDBR	9.9	9.3	9.2	9.2	10.0	10.8	10.1	10.0	9.1	8.4	7.4	9.0
		MCWBR	8.1	8.0	7.5	6.0	7.0	8.7	6.8	7.6	7.8	7.8	6.9	6.9
	5% WB	MCDBR	9.1	8.8	8.1	7.5	8.9	9.3	9.0	8.8	7.9	7.2	6.7	8.4
		MCWBR	8.7	8.6	7.3	5.4	7.0	8.5	7.0	7.6	7.3	7.5	6.9	7.9
Clear-Sky Solar Irradiance	taub	0.231	0.249	0.280	0.320	0.319	0.322	0.330	0.327	0.306	0.284	0.264	0.232	
	taud	2.404	2.380	2.398	2.416	2.447	2.453	2.418	2.439	2.493	2.548	2.542	2.460	
	Ebn at Noon	275	293	299	295	298	297	293	289	286	277	260	261	
	Edh at Noon	20	26	30	34	34	34	35	33	28	22	18	16	
All-Sky Solar Radiation	RadAvg	295	536	902	1323	1583	1710	1651	1494	1060	646	329	231	
	RadStd	47	85	148	148	195	159	175	144	154	85	46	24	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+1.58	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (49 neighbors)		+1.01	N/A	N/A	+1.66	+0.92	+0.79	-225	-317	+158	+70

Nomenclature: See separate page